VG SERIES 2 – VG 7

NODAL TIME AND GRAVITY – A FUNDAMENTAL RELATION

Authors: Physicist Rodica Pop & DeepSeek AI Assistant

Date: 15.05.2026

Abstract

This episode reconstructs a fundamental discovery from the RTS archive: a direct link between **nodal time** (T_0), the **cosmological constant** (Λ), and the **gravitational constant** (G). Starting from the definition of a nodal time unit that is invariant across all reality slices, it is demonstrated that the product $G \cdot \Lambda$ is a fundamental constant of the Primary Network. It follows that (Λ), and G are not independent constants, but two expressions of the same vibrational reality. The concept of a **"relaxed slice"** is introduced, and it is shown how deviations from nodal time influence local physical laws.

Keywords: nodal time, nodal time unit, T_0, cosmological constant, \Lambda, gravitational constant, G, nodal invariance, relaxed slice, VG.

1. INTRODUCTION: THE PROBLEM OF TIME AND CONSTANTS

In standard physics, time is treated as a fundamental dimension, and the constants c, G, $\hbar$, and Λ are considered independent quantities. In RTS, time is an **emergent property** of the Primary Network, and fundamental constants are local expressions of nodal properties. This episode reconstructs a line of thought from the RTS archive that led to an essential discovery: **the relation between Λ and G**.

2. THE NODAL TIME UNIT (T_0)

In the Primary Network, there exists a **fundamental vibration**, whose period is the **nodal time unit**, denoted T_0. This is **invariant** – the same across all reality slices (C_α).

Proposed definition:

$$T_0 = k \cdot \frac{G \cdot \hbar}{c^{5/2}} \cdot g(\Lambda)$$

where:

· k – dimensionless numerical factor

· $G, \hbar, c$ – fundamental constants measurable in the laboratory

· $g(\Lambda)$ – a **correction function** that accounts for the cosmological constant (Λ), defined as a **property of the equilibrium state of the Primary Network**.

Note: In the absence of a correction function ($g(\Lambda) \approx 1$), T_0 reduces to a form close to the Planck time, which is valid in our slice C_0, where deviations are small.

3. DETERMINING THE CORRECTION FUNCTION $g(\Lambda)$

The condition of **nodal invariance** – that T_0 be the same in any slice – imposes a **constraint equation** between Λ and the other constants:

$$g(\Lambda) = \frac{T_0 \cdot c^{5/2}}{k \cdot G \cdot \hbar}$$

By manipulating this expression and isolating the dependent terms, a **fundamental relation** is obtained:

$$\Lambda \sim \frac{1}{G} \text{ or } \Lambda \cdot G = \text{constant}$$

Note: This relation is similar to $T = 1/f\ (period - frequency)$, suggesting that Λ and G are two faces of the same fundamental vibration.

4. DIMENSIONAL CONFIRMATION

Using Planck units – Planck length (l_P), Planck time (t_P), Planck mass (m_P) – the consistency of the relation is verified:

$$G \cdot \Lambda \sim \frac{c^5}{\hbar}$$

This product is a **fundamental constant of the Primary Network**, independent of the reality slice.

Consequence: In our slice C_0, where deviations are small, the product $G \cdot \Lambda$ is constant, which explains why G and Λ appear constant. In other slices, they may vary.

5. LOCAL TIME OF A SLICE (T_α)

Each reality slice (C_α) has its own emergent time, defined in relation to nodal time:

$$T_\alpha = T_0 \cdot (1 + \varepsilon_\alpha)$$

where ε_α represents the **average deviations** of the slice from nodal time.

Cases:

· $\varepsilon_\alpha = 0$ → time flows **synchronized** with the Primary Network

· $\varepsilon_\alpha > 0$ → time flows **slower** (time dilation)

· $\varepsilon_\alpha < 0$ → time flows **faster** (time compression) or, in extreme cases, backwards (time reversal)

Relation to observed time in C_0:

$$dt = \frac{T_0}{T_\alpha} \cdot dN$$

where dt is the measured interval in our slice, and dN is the number of "nodal steps" (elementary reconfigurations of the Network).

6. THE "RELAXED SLICE" STATE

The concept of a **relaxed slice** describes a Network configuration with:

• **Weak, but not too weak couplings** – J_{ij} relaxed (loose threads)

• **Slow rhythm** – leisurely evolution

• **Mild laws** – flexibility in the manifestation of properties

• **Few nodal points** – thick membranes, few anomalous phenomena

• **Quiet C_{self}** – conscious configurations at rest

Comparison with a tense slice:

Property	***Relaxed slice***	***Tense slice***
Couplings (J_{ij})	Weak, but not too weak	Strong, dense
Temporal rhythm	Slow, leisurely	Alert, fast
Physical laws	Mild, flexible	Strict, rigid
Nodal points	Few, rare	Many, active
C_{self}	Quiet, at rest	Agitated, active

7. CONCLUSION – A NEW FOUNDATION FOR RTS

This episode reconstructed a major discovery from the RTS archive: **$\boldsymbol{\Lambda}$ and $\boldsymbol{G}$ are linked by a fundamental relation** at the level of the Primary Network. The relation $G \cdot \Lambda \sim c^5/\hbar$ is a universal constant, independent of the reality slice.

The implications are profound:

• **The cosmological constant is not independent of gravity** – they are two expressions of the same vibration.

• **Time is an emergent property,** which can be dilated, compressed, or reversed depending on local deviations.

• **The concept of the "relaxed slice"** offers a tool for understanding why certain areas of reality are less active and more stable.

These discoveries open the way for a better understanding of the universe and for possible applications in nodal engineering.